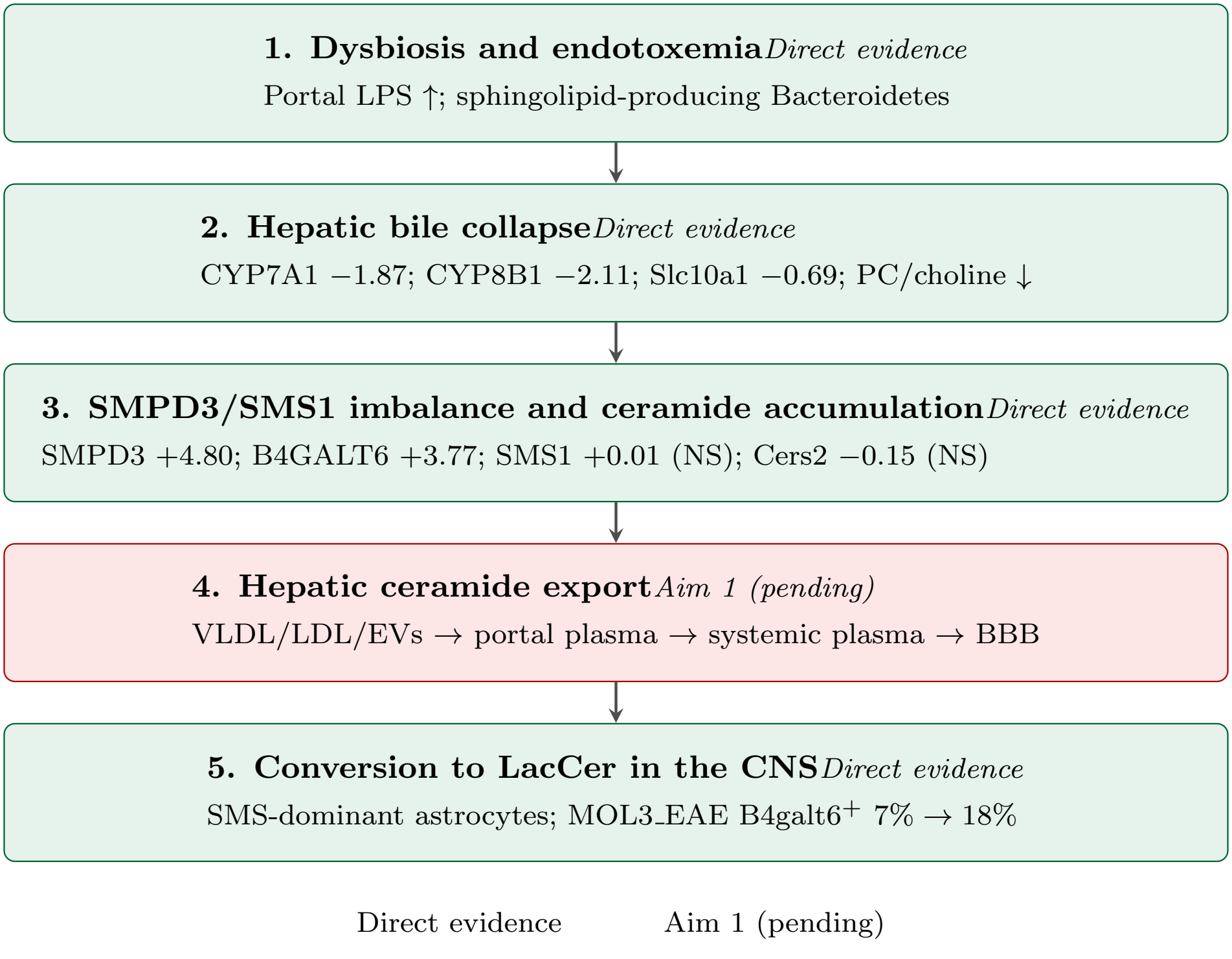

1. Dysbiosis and endotoxemia Direct evidence
Portal LPS ↑; sphingolipid-producing Bacteroidetes
2. Hepatic bile collapse Direct evidence
CYP7A1 −1.87; CYP8B1 −2.11; Slc10a1 −0.69; PC/choline ↓
3. SMPD3/SMS1 imbalance and ceramide accumulation Direct evidence
SMPD3 +4.80; B4GALT6 +3.77; SMS1 +0.01 (NS); Cers2 −0.15 (NS)
4. Hepatic ceramide export Aim 1 (pending)
VLDL/LDL/EVs → portal plasma → systemic plasma → BBB
5. Conversion to LacCer in the CNS Direct evidence
SMS-dominant astrocytes; MOL3_EAE B4galt6+ 7% → 18%
Direct evidence
Aim 1 (pending)